

# Week 1 Report: Understanding Vision Transformers and DINOv2 for DESI Galaxy Images

## 1. Why ViT?

Vision Transformer (ViT) is a way of applying the Transformer idea to images. Instead of processing an image using convolution filters like a CNN, ViT breaks the image into small patches and treats those patches like tokens.

This is useful for our project because we want to understand the visual representation learned by a modern pretrained vision model for galaxy images. ViT also gives us a clear way to look at both the overall image representation and the representation of individual image regions.

A CNN has stronger built-in assumptions about images, such as locality. ViT has fewer image-specific assumptions and learns relationships between image patches using self-attention.

## 2. Patchification

Patchification simply means cutting an image into small fixed-size pieces called patches.

For our experiment, DINOv2 uses a patch size of  $14 \times 14$  pixels. Before entering the model, our DESI RGB image was processed from  $160 \times 160$  to  $224 \times 224$  pixels.

$224 / 14 = 16$  patches along each direction, so the image gives  $16 \times 16 = 256$  patches.

Each of these patches becomes one token for the Transformer.

## 3. Patch Embeddings

The patches are converted into numerical vectors called patch embeddings. In DINOv2-S/14, each patch is represented using 384 numbers. Therefore, we have 256 patch embeddings, each with 384 features.

We can think of this as 256 patches  $\times$  384 features. These numbers are learned representations. We should not assume that one particular number corresponds to one specific visual property; the meaning is distributed across the whole embedding.

## 4. Positional Embeddings

Once the image is converted into patches, the Transformer needs information about where each patch came from. For example, a patch from the centre of a galaxy and a patch from the corner may contain similar patterns, but their positions are different.

Positional information helps the model keep track of the spatial arrangement of the patches. DINOv2 already handles this as part of the pretrained model, so we did not implement positional embeddings ourselves.

## 5. CLS Token

The CLS token is a special token added to the sequence of image patch tokens. It is intended to contain a global representation of the image after passing through the Transformer.

Our image produced 256 patch tokens. With one CLS token, the total becomes 257 tokens.

The model output was [1, 257, 384], where 1 means one image, 257 is the number of tokens, and 384 is the feature dimension. We extracted the first token as the CLS token and obtained [1, 384].

## 6. ViT Transformer Architecture

A Vision Transformer takes the sequence of image tokens and passes them through Transformer blocks. The important part is self-attention. Self-attention allows one token to interact with other tokens, so a patch can use information from other parts of the image.

A Transformer block contains Multi-Head Self-Attention (MSA) and an MLP, along with LayerNorm and residual connections. Our DINOv2-S/14 model printout showed 12 Transformer layers, a feature dimension of 384, and an MLP that expands the features from 384 to 1536 and then back to 384.

The simple idea is: self-attention mixes information between tokens, while the MLP transforms the features of each token.

## 7. CLS vs Patch Tokens

The CLS and patch representations serve different purposes.

CLS token: [1, 384] — one compact representation of the overall image.

Patch tokens: [1, 256, 384] — 256 representations corresponding to the image patches.

A simple way to remember it is: CLS asks, 'What does the whole image look like?' Patch tokens ask, 'What representation does the model have for each part of the image?' This is useful because galaxy morphology can depend on both global structure and local features.

## 8. What is DINOv2?

DINOv2 is a pretrained computer vision model based on the Vision Transformer architecture. It uses self-supervised learning, meaning the model can learn useful visual representations without requiring a manually provided class label for every image during pretraining.

We used the pretrained model facebook/dinov2-small, which in our experiment is DINOv2 ViT-S/14.

## 9. What does DINOv2 output?

For our DESI galaxy image, DINOv2 produced [1, 257, 384]. This means one image, 257 tokens, and 384 features per token.

The 257 tokens consist of 1 CLS token and 256 patch tokens. We extracted them separately: CLS token → [1, 384], patch tokens → [1, 256, 384].

We also calculated the magnitude of each patch embedding and reshaped the 256 values into a 16×16 grid. This gave us a simple spatial visualization of the patch representations.

## 10. How will DINOv2 be used in this project?

The main purpose of using DINOv2 is to obtain useful visual features from galaxy images. Instead of starting with raw pixels and immediately building a classifier, we first use DINOv2 to convert each galaxy image into embeddings.

The CLS embedding provides a compact global representation, while the patch embeddings provide more detailed spatial information. These representations can later be used for exploring galaxy similarity, morphology-related structure, or as features for a downstream model.

the main goal is to understand the data, understand how ViT/DINOv2 represents the image, and verify that we can successfully extract and inspect the representations.

## 11. Dataset Exploration and Galaxy Zoo Labels

The dataset used was MultimodalUniverse/legacysurvey. A sample contained image data, masks, inverse variance information, PSF information, scale information, catalog measurements, and object-level metadata.

The image section contained four bands: des-g, des-r, des-i and des-z. Each flux image was 160×160.

We also inspected catalog fields such as FLUX\_G, FLUX\_R, FLUX\_I, FLUX\_Z, TYPE, SHAPE\_R, SHAPE\_E1, SHAPE\_E2, X and Y.

An important observation is that the inspected dataset schema did not contain explicit Galaxy Zoo morphology vote/label columns. Therefore, TYPE should not be presented as a Galaxy Zoo morphology label. If Galaxy Zoo labels are needed later, they need to be obtained from the relevant Galaxy Zoo catalogue or cross-match.

## 12. Practical Workflow and Results

The practical workflow was: DESI Legacy Survey sample → inspect g/r/i/z images → use the provided RGB image → preprocess it with the DINOv2 processor → obtain a 224×224 RGB tensor → run the pretrained DINOv2 model → obtain [1, 257, 384] → separate CLS and patch tokens → visualize patch embedding magnitudes.

Main results:

Input to DINOv2: [1, 3, 224, 224]

Model output: [1, 257, 384]

CLS token: [1, 384]

Patch tokens: [1, 256, 384]

Patch grid: [16, 16]

## 13. What I Understood from Week 1

The main thing I understood is that a Vision Transformer does not look at an image exactly like a CNN. It first converts the image into patches and treats those patches as tokens.

DINOv2 then uses a Transformer to build richer representations of those tokens. The CLS token gives a compact representation of the complete image, while the patch tokens retain information from different parts of the image.

For our galaxy project, this gives us a useful starting point: instead of working only with raw pixels, we can work with learned visual features and later investigate whether these features contain information related to galaxy morphology.